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Cancer incidence patterns in Koreans in the US and in Kangwha, South Korea

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Abstract

Objective: In the US, Koreans are a rapidly growing group and comprised 10.5% of the total Asian population as of 2000. However, little has been published regarding cancer patterns in this subpopulation.
Methods: Using data from the Surveillance, Epidemiology, and End Results program, the California Cancer Registry, and the International Association for Research on Cancer, we compared age-adjusted and age-specific incidence rates for cancers of the prostate, breast, cervix, lung, colon, rectum, stomach, liver, and esophagus in US Koreans with rates of these cancers in residents of Kangwha, South Korea, and in US whites as a reference.
Results: While the most frequently diagnosed cancer was lung among US Korean males and breast among US Korean females, it was stomach cancer for both sexes in Kangwha. Rates of prostate, breast, and colon cancer were considerably higher for Koreans in the US than in Kangwha, but were not as high as in whites. Cervical and stomach cancers showed the opposite racial/ethnic pattern, with rates highest in Kangwha, intermediate among US Koreans, and lowest among whites. Rates of rectal cancer in females and esophageal cancer in males were two-times higher in Kangwha than in US Koreans but esophageal cancer rates were similar between US Koreans and whites. Liver cancer rates were similar between Kangwha residents and US Koreans, but nearly 10-times lower among whites.
Conclusions: Although these comparisons may have methodologic limitations, including data quality and racial/ethnic misclassification, the differences seen in migrant and native Koreans for some cancers warrant further investigation in this growing subpopulation.

Introduction

Some of the most dramatic inter-population differences in cancer occurrence are observed between native populations in Asia and whites in the United States (US) [1–4]. Thus, data from epidemiologic studies of migrants from Asia to the US offer a particularly interesting resource for the generation of hypotheses regarding possible genetic and environmental contributions to cancer etiology [5–8].

The US has continually attracted migrants from diverse social, political, and economic backgrounds into its great 'melting pot' [9]. In 2000, over a quarter of the estimated 28.4 million foreign-born people in the US were migrants from Asia [10]. Koreans, numbering 1.2 million according to the 2000 Census, are the fifth largest Asian group in the US, comprising 10.3% of the US Asian population [11]. The first immigrants in this group, mostly male bachelors, came to the US from Korea in the late 1800's and early 1900's, settling in Hawaii as plantation laborers [12]. The second wave of Korean immigration, which occurred after the Immigration Act of 1965 repealed national-origins quotas, primarily consisted of women coming to marry Korean–Americans ('picture brides'), American military brides, war-orphaned children adopted by Americans, and students [13]. Because the second wave was the larger, 73% of the US Korean population was foreign-born as recently as 1990 [14]. Although the second wave included more college-educated middle-class professionals than the first, many Korean professionals, like other Asian migrants, ended up working in jobs less skilled

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than their training [12]. Census data show that, in 1990, 20% of recent immigrants were living in poverty, compared to 8% of US-born and 7–10% of foreign-born early migrants [14]. There were also striking gender differences in socioeconomic status; a considerably larger proportion of females, particularly recent migrants, than males lacked a high school education (about 25% among females and 10% among males) [14]. The proportion of US Koreans with poor English-speaking skills correlated directly with time since immigration, with 40% of recent immigrants speaking poor or no English. These three characteristics indicate that, in general, among Koreans, recent immigrants were of relatively lower socioeconomic status than the US-born or early immigrants.

Despite growing presence and socioeconomic diversity of Koreans in the US, little has been published regarding their cancer patterns and how they have changed with migration. While prior systematic reviews of cancer incidence have compared Chinese, Japanese, and Filipinos in the US to their counterparts in Asia [15–30], similar incidence comparisons have not been presented for Koreans. Therefore, we compared cancer incidence rates for Koreans living in the US to rates for residents of Kangwha County, South Korea, as well as to rates for US whites, for cancer sites with substantial burden in Korean populations.

Materials and methods

For the US populations, we obtained incidence data on invasive cancers for the period 1988–1992 from the statewide California Cancer Registry and from the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) program, which cover the states of Connecticut, Hawaii, Iowa, New Mexico, and Utah, and the metropolitan areas of Atlanta, Detroit, and Seattle/Puget Sound (Los Angeles, San Francisco/Oakland, and San Jose/Monterey registries are also part of SEER, but are included in the California data) [31, 32]. The catchment regions of SEER and the California Cancer Registry together covered 43% of the 1990 US Korean population [32]. The California Cancer Registry and SEER provided mid-year population estimates by sex, race/ethnicity, and age based on the 1990 US Census. For the California Cancer Registry and other registries comprising SEER, cancer registration is compulsory, and data completeness is estimated to be >97% [3, 25].

Kangwha County is a rural island in South Korea 50 km from Seoul, with a mostly agricultural population [3]. For this area, incidence data on invasive cases and population estimates were obtained from the International Association for Research on Cancer's (IARC) Cancer in Five Continents for the period 1986–1992; this was the only registry in Korea that reported data to IARC [3]. Cancer reporting in this region is voluntary, and cases diagnosed in outpatient clinics or by autopsy were not reportable [3]. In addition, death registration is not required; thus, cancers diagnosed at the time of death and not reported by another source may have been missed [3]. Nevertheless, registries selected to be included in IARC are required to maintain a certain level of data quality and comparability to other registries. The criteria for inclusion include the definition of an incident case of cancer, the completeness of enumeration of cases in the population covered, the accuracy of abstraction and coding of information, and the accurate estimation of the denominator [3].

To quantify the relative burden caused by specific cancers, we computed the proportion of all case counts contributed by the five most common cancer sites for US non-Hispanic whites (hereafter called 'whites'), US Koreans, and Koreans living in Kangwha [24], with race/ethnicity in US data abstracted from patients' medical records or death certificates. We also calculated average annual rates per 100,000 age-adjusted to the 1970 world standard population [3] using the direct method [33] and excluding cases of unknown age, and 95% confidence intervals based on the gamma distribution [34]. Age-specific rates were computed for five age groupings and were not adjusted for age. We also present ratios of age-adjusted incidence rates for US whites and US Koreans, and for US Koreans and Koreans in Kangwha.

Results

Prostate cancer

While prostate cancer was the largest contributor to the overall cancer incidence burden among US whites (29%), it comprised only 7% of all cancer cases diagnosed among US Koreans (Figure 1). The incidence rate of prostate cancer in US whites was nearly six-times that of US Koreans, while the rate in Kangwha males was negligible (Table 1). Age-specific data indicate that the incidence differences between US whites and Koreans occurred mainly in men age 65 and older (Figure 2).

Breast cancer

Breast cancer comprised 31% of cancers among US whites and 19% among US Koreans, making it the most

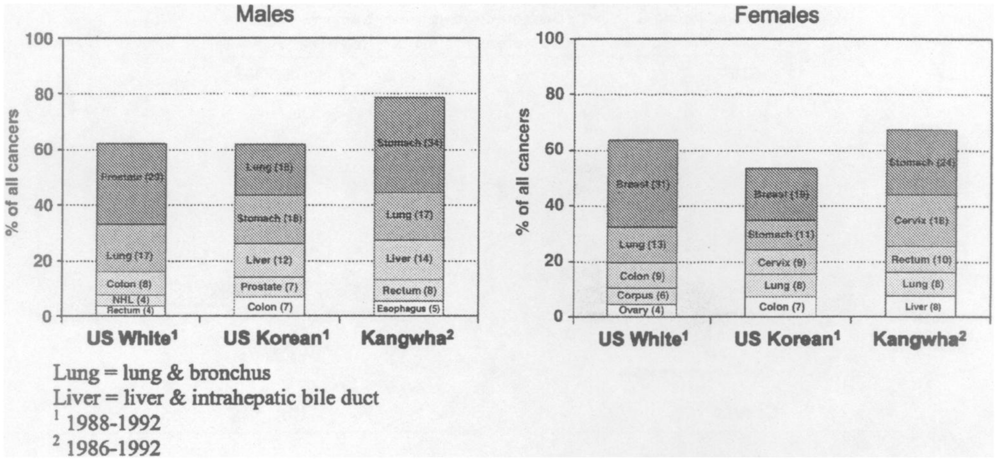

Fig. 1. Five cancers contributing most to overall cancer incidence burden, by racial/ethnic group, sex, and region.

Table 1. Average annual age-adjusted incidence rates[a] and 95% confidence intervals (CI), and incidence rate ratios, by cancer site, race/ethnicity, region, and sex

Cancer site Sex	Non-Hispanic white (US[b])			Korean						Rate ratio	
				US[b]			Kangwha, Korea[c]			White: US Korean	US Korean: Kangwha
	Count	Rate	95% CI	Count	Rate	95% CI	Count	Rate	95% CI		
Prostate	123,775	101.0	100.4–101.6	93	17.2	13.8–20.9	< 5	[d]	[d]	5.9	[d]
Breast (female)	122,871	98.7	98.1–99.3	261	25.1	22.1–28.4	23	7.1	4.5–10.4	3.9	3.5
Cervix	7993	7.3	7.2–7.5	125	13.0	10.8–15.5	71	21.8	16.9–27.3	0.6	0.6
Lung[e]											
Male	71,680	64.4	63.9–64.8	232	40.8	35.6–46.4	103	33.1	26.9–39.8	1.6	1.2
Female	50,546	37.7	37.4–38.1	113	12.7	10.5–15.2	32	8.4	5.7–11.7	3.0	1.5
Colon											
Male	34,594	29.1	28.7–29.4	85	14.3	11.3–17.6	14	4.5	2.4–7.1	2.0	3.2
Female	36,467	21.4	21.1–21.6	97	10.7	8.6–13.0	7	2.3	0.9–4.3	2.0	4.7
Rectum											
Male	15,163	13.4	13.2–13.6	68	11.1	8.5–14.0	46	15.4	11.3–20.3	1.2	0.7
Female	12,123	8.0	7.8–8.1	53	5.4	4.0–7.0	37	10.0	6.9–13.5	1.5	0.5
Stomach											
Male	8347	7.2	7.1–7.4	223	36.5	31.7–41.6	203	66.7	57.8–76.4	0.2	0.5
Female	4938	2.9	2.8–3.0	148	15.7	13.3–18.4	91	25.1	20.1–30.7	0.2	0.6
Liver[f]											
Male	3089	2.8	2.7–2.9	150	23.6	19.9–27.6	85	28.5	22.7–35.0	0.1	0.8
Female	1730	1.2	1.1–1.2	77	8.6	6.8–10.7	29	7.7	5.1–10.8	0.1	1.1
Esophagus											
Male	4708	4.3	4.2–4.4	26	4.6	3.0–6.6	32	10.4	7.1–14.3	0.9	0.4
Female	2059	1.3	1.3–1.4	< 5	[d]	[d]	< 5	[d]	[d]	4.3	[d]

[a] Rates are per 100,000, adjusted to the 1970 world standard population.
[b] US = SEER + California, 1988–1992.
[c] Kangwha County, Korea, 1986–1992.
[d] Data are not shown for rates based on fewer than 5 cases.
[e] Lung = lung and bronchus.
[f] Liver = liver and intrahepatic bile duct.

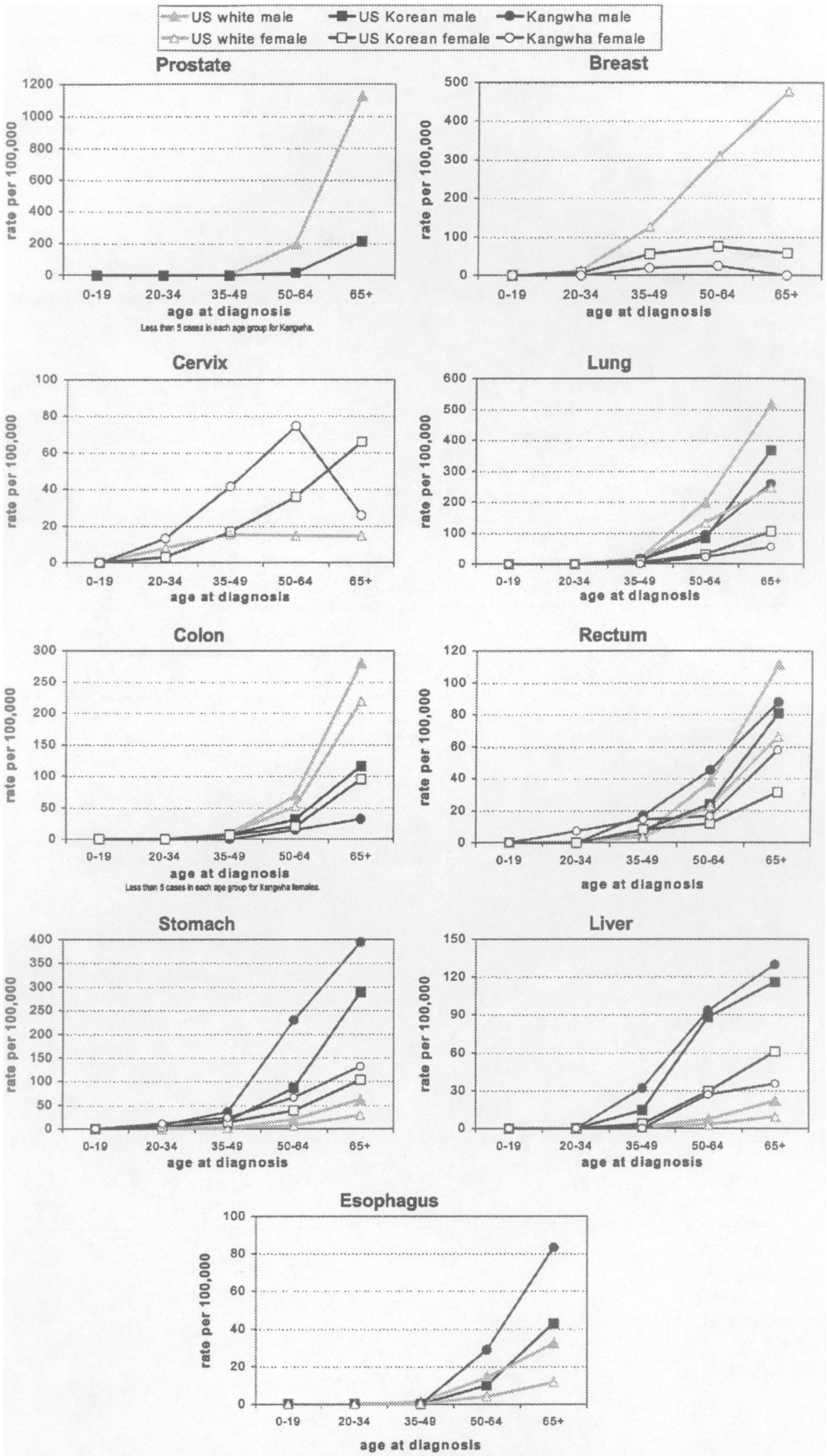

Fig. 2. Age-specific incidence rates per 100,000 by cancer site, race/ethnicity, region, and sex (1988–1992 for US, 1986–1992 for Kangwha).

common cancer in both racial/ethnic groups (Figure 1). However, among females in Kangwha, it did not rank among the five most common cancers. Incidence rates (Table 1) were almost four-times higher for US whites than US Koreans, and 3.5-times higher among Koreans in the US than in Kangwha. Age-specific incidence curves (Figure 2) revealed marked variations between whites and US Koreans, particularly after age 50, but patterns were similar for Koreans in the US and Kangwha, with no exponential increase after age 35, as seen for US whites, and even decreasing rates after age 64 in both groups.

Cervical cancer

In Kangwha, cervical cancer was the second most commonly diagnosed cancer and represented 18% of overall cancer incidence, compared to 9% among US Korean females (Figure 1). The cervical cancer rate in Kangwha was approximately double that of US Koreans, and three-times that of US whites. Among Kangwha females, rates peaked in the 50–64 age group to a level that was double that of US Koreans, but subsequently declined to a level that was closer to US Whites. Rates among US women in the ages 35–49 were similar for Koreans and whites, but with age increased steadily among Koreans, while remaining constant among whites (Figure 2).

Lung cancer

As seen in Figure 1, lung cancer was among the strongest contributors to the overall incidence burden for Korean males irrespective of geography. Among Korean females, lung cancer constituted 8% of cancers both in the US and Kangwha. However, rates were 1.6-times higher for US white than US Korean males, and three-times higher for US white than US Korean females (Table 1). Rates were slightly higher in US Koreans than their counterparts in Kangwha, 1.2-times for males, and 1.5-times for females. Age-specific rates were consistent with the age-adjusted patterns (Figure 2).

Colon and rectal cancer

Colon cancer rates were approximately double rectal cancer rates and exhibited greater variation between US whites and US Koreans (Table 1). In Kangwha, however, rectal cancer rates were higher than colon cancer rates. Thus, while colon cancer was the fifth most common cancer among US Korean males and females (Figure 1), it did not rank among the five most common cancers among Kangwha males or females; rather, rectal cancer exceeded colon cancer in incidence and was the fourth and third most common cancer among Kangwha males and females, respectively. Age-specific rates (Figure 2) show that the variations were greatest in the oldest age group.

Stomach cancer

Stomach cancer was the most common cancer in Kangwha among both males and females (Figure 1). Table 1 shows that rates for whites were one-fifth those for US Koreans. Stomach cancer remained a significant burden among US Koreans (Figure 1); nevertheless, their rates were half those of residents in Kangwha (Table 1). Figure 2 shows that age-specific rates started to diverge among the population groups after age 49.

Liver cancer

Primary liver cancers were among the five most common cancers diagnosed among Korean males in the US and Kangwha (Figure 1). Liver cancer exhibited marked gender and ethnic variations; incidence rates were two- to four-times higher among males than females and were substantially greater (10-fold) among Koreans than whites, regardless of geography (Table 1). The incidence of liver cancer increased steadily after age 49 (Figure 2), without substantial variation among racial/ethnic groups, regardless of their region of residence.

Esophageal cancer

Esophageal cancer, though rare among whites in the US, was the fifth most commonly diagnosed cancer of males in Kangwha (Figure 1). Among US Korean males, the incidence rate was similar to that for US whites and less than half that for Kangwha males (Table 1). Age-specific rates were most variable among the population groups age 65 and above, such that rates among Kangwha males were twice those of US Korean males (Figure 2).

Discussion

Our comparisons of incidence rates of nine common cancers for Koreans and whites living in the US to rates for residents of Kangwha County, South Korea, have revealed several noteworthy patterns suggestive of etiologic clues. One pattern involves cancers for which rates in US whites are higher than rates in US Koreans, which in turn are higher than rates in Kangwha; this pattern includes cancers of the prostate, breast, colon, and to some extent, lung. A second pattern, the reverse of the first, is seen for cervical and stomach cancers; these diseases are most common in Kangwha, intermediate in US Koreans, and less common in US whites. As the majority of Koreans living in the US are foreign-born, these first two patterns suggest that the associated cancers are influenced in part by migration-related changes in risk factors. However, prostate, breast, and cervical cancer rates can also be influenced by screening, a factor that can complicate the interpretation of rate differences. A third pattern is seen for liver cancer, for which rates are equal among US Koreans and Koreans in Kangwha but considerably lower among US whites. This pattern suggests that migration-related changes in environmental factors do not affect disease pathogenesis, and that exposure changes over more than one generation may be required for changes in risk of this disease; alternatively, US Koreans retain the same level of liver cancer risk factors as do Koreans in Asia. Given that the primary risk factor for liver cancer is infection with the hepatitis B and C viruses, the incidence patterns may be reflective of geographical variations in the prevalence of these viruses, or in promotional risk factors for liver cancer. A fourth pattern is seen for rectal cancer in females and esophageal cancer in males, for which rates are higher in Kangwha than among US Koreans, suggesting that the pathogenesis of these cancers is influenced immediately by changes in environment. The major risk factors that have been found to be associated with esophageal cancer include alcohol and tobacco use, and consumption of hot beverages, pickled vegetables, and foods containing nitrites and nitrosamines [25]. The dietary risk factors, in particular, are highly prevalent in traditional Korean culture, and may decrease in prevalence upon migration to the US. With regards to rectal and colon cancer, the differences in incidence patterns support different etiologies and risk factors for these two sites that are traditionally combined together in studies.

The vast differences in social class by migrant status among the US Korean population suggest that there may be further variations in cancer risk for this group, particularly for sites with strong associations with socioeconomic status, such as breast and cervical cancers. For example, the age-specific incidence pattern of cervical cancer among the Korean populations may be attributable to underutilization among older Korean females of Pap smears, which detect cervical tumors at a pre-invasive stage. Low cervical cancer screening has been associated with lower education and lack of a regular health care provider [35]; these characteristics are more prevalent among recently migrated than US-born Koreans, suggesting that the relatively higher cervical cancer rates seen among US Koreans may be occurring primarily in immigrant Korean women. Similarly, breast cancer occurs more often among women of higher than lower socioeconomic status, suggesting that rates may be even higher among early migrants than recent migrants.

The implementation of cancer screening differ not only internationally but also among racial/ethnic groups in the US, with lower screening practices noted for Koreans than whites [35–42]. For example, 25 and 50% of Korean women reported never having received a Pap smear or mammogram, compared to 5 and 21%, respectively, of white women [38]. Differences in screening practices among populations can complicate interpretations of incidence rate ratios. In the US, the population-wide implementation of screening for breast and prostate cancers has resulted in dramatic peaks and subsequent declines in incidence rates [43–47]. These effects of screening may be particularly relevant for the ethnic differences in prostate incidence, as the US data we examined captured a time period that included the previously described rate peak among whites in 1992 but did not include the later incidence peak noted for Asians in 1993 [48, 49]. The question is whether the observed incidence rate ratios are attributable to screening *per se* or to other aspects of lifestyle and dietary changes upon migration.

Methodological issues arise when comparing incidence rates in migrant populations in the host country (*e.g.*, US) with those in their native country (*e.g.*, South Korea). An inherent concern of migrant studies is a type of selection bias commonly called the 'migrant effect' [8], in which migrants may not be directly comparable to the population from which they migrated.

In addition, for this study, the US Korean population may differ from the population in Kangwha County, an agricultural region, especially as US Koreans have predominantly migrated from more urban regions in South or North Korea [12]. This difference would be mostly relevant for those cancers with documented urban-rural effects on risk, such as breast and prostate cancers [50]. Differences in incidence rates may also be attributable to differences in cancer detection, particularly in Kangwha, where cancer screening and early detection are unlikely to be as thorough as in the US. Relatedly, data quality and comparability across cancer registries may be affected by completeness in case ascertainment and diagnostic accuracy [3]. This is a particular concern for Kangwha, a newly established registry in a region where cancer registration is voluntary and ascertainment from certain data sources, such as death certificates, may be incomplete [3]. Thus, incidence rates in Kangwha for relatively fatal cancers (*e.g.*, lung, liver, and stomach cancers) may be underestimated as a result of the poor mortality data. However, in the US, the proportion of lung, liver, and stomach cancers diagnosed at death is <1% [51]; although this proportion may be higher in Kangwha, presumably most of the cases missed by death certificates will have been reported by other sources, such as hospital records. While international comparisons may be hampered by data quality, we feel that it is nevertheless worthwhile to present these data, given that no such comparisons have been conducted.

Incidence rates in the US may be biased by misclassification of persons by race/ethnicity. We previously documented such misclassification between Vietnamese and Chinese patients in the San Francisco/Oakland cancer registry and a consequent overestimate of incidence rates in Vietnamese [52]. Similar misclassification may affect other US Asian subgroups, but no studies, to our knowledge, have evaluated this error in Koreans. Estimates of the population at risk, or the denominator, also may be subject to uncertainty. US population counts provided by the Census are the best available denominator data, but limitations such as the undercounting of minority groups in the 1990 Census exist [53]. The quality of the population counts in Kangwha is unknown.

Despite these methodologic issues, the international patterns observed in our paper are consistent with those using data from Asian countries with better-established cancer registries [25]. Population-based comparisons between migrants from specific Asian subgroups and their native populations are particularly valuable given the tendency of health research to assess US Asians as one heterogeneous group. US Koreans have different immigration histories and dietary and lifestyle practices from other Asian groups. We have also seen that US Koreans have a cancer profile that is unique from non-Hispanic whites in the US and Koreans in Kangwha. Oversampling Koreans for epidemiologic studies and considering them separately from other Asian subgroups could help produce further insights into the relative contributions of heredity and environment to cancer development and their potential gene-environment interactions, as well as provide needed information regarding disease etiology and cancer control strategies specific to this dynamic population group.

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